

Back to our problem. Going through  $y = x$  is the same as intersecting with  $y = x - 1$  in this case, so  $(0, 0)$  can be reflected to  $(1, -1)$ , and the number of intersecting paths is  $\binom{m+n}{m-1}$ . Therefore, the answer to the problem is

$$\binom{m+n}{m} - \binom{m+n}{m-1} = \frac{n-m+1}{n+1} \binom{m+n}{m}$$

In particular, when  $n = m$ , the formula becomes

$$\frac{1}{n+1} \binom{2n}{n}$$

which is the **Catalan number**.

▼ Note

Note that there are two ways to generate valid paths in the first description when  $n = m$ , one keeps  $x \geq y$  and the other keeps  $x \leq y$ , so the answer is  $\frac{2}{n+1} \binom{2n}{n}$ .